A strict numerical-index bound for real Lebesgue spaces

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Abstract

We show that the numerical index of a real Lebesgue space is strictly below the proposed two-dimensional value at every non-Hilbert exponent between one and infinity. A three-dimensional analytic perturbation proves the general strict inequality. Independent exact algebraic and rational examples give explicit certificates at exponent four and its conjugate.

Repository target. This manuscript concerns AIM problem 935 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

Answer. The proposed identity is false for every $1 < p < \infty$ with $p \neq 2$. A three-dimensional perturbation gives the strict inequality throughout this range. An independent rational certificate at $p = 4$ supplies a concrete matrix and an explicit numerical gap; its transpose also treats $p = 4/3$.

1.1 Strict inequality for every non-Hilbert exponent

The counterexample admits an analytic extension to every exponent in the question. The explicit rational certificate below remains useful because it supplies a concrete matrix and a numerical gap at $p = 4$.

Theorem 1.1. *For every $1 < p < \infty$ with $p \neq 2$,*

$$n\big(L_p([0,1];\mathbb{R})\big) \leq n\big(\ell_p^3(\mathbb{R})\big) < M_p.$$

In particular, the proposed equality fails for every exponent to which Problem 935 applies.

We prove the theorem for $p > 2$ by perturbing an operator that attains the candidate ratio. The perturbation changes its numerical radius by at most $O(\varepsilon^2)$, while increasing a lower bound for its norm by a positive multiple of $|\varepsilon|^{p/(p-1)}$. The latter exponent is strictly smaller than two.

Lemma 1.2. *For $p > 2$, the function*

$$f_p(t) = \frac{t - t^{p-1}}{1 + t^p}, \qquad 0 \leq t \leq 1,$$

has a unique maximum at some $t = t_p \in (0,1)$, and $f_p''(t_p) < 0$. Set $j_p(s) = |s|^{p-2}s$ for real s, with $j_p(0) = 0$. The function

$$h_p(s) = \frac{j_p(s) - s}{1 + |s|^p}$$

has maximum M_p precisely at $s = t_p^{-1}$ and $s = -t_p$, and minimum $-M_p$ precisely at $s = t_p$ and $s = -t_p^{-1}$. Each of these extrema is nondegenerate.

Proof. Write $t = e^{-x}$, $a = (p-2)/2$, and $b = p/2$. For $x > 0$,

$$f_p(e^{-x}) = \frac{\sinh(ax)}{\cosh(bx)}.$$

Its logarithmic derivative is

$$a\coth(ax) - b\tanh(bx),$$

whose derivative is

$$-a^2\operatorname{csch}^2(ax) - b^2\operatorname{sech}^2(bx) < 0.$$

The logarithmic derivative tends to $+\infty$ as $x \downarrow 0$ and to $a - b = -1$ as $x \to \infty$. It therefore has exactly one zero, where the second derivative of the positive function is strictly negative. The change of variable transfers nondegeneracy to t_p. Both endpoint values of f_p are zero, so this is its unique global maximum. The assertions about h_p follow from its oddness and the relations $h_p(s) = -f_p(s)$ for $0 < s < 1$ and $h_p(s) = f_p(1/s)$ for $s > 1$. □

Proof of Theorem 1.1 for $p > 2$. Fix $p > 2$, write $q = p/(p-1)$, and put

$$c = 2^{-1/p}, \qquad u = (c, c), \qquad w = (c, -c).$$

Thus $\|u\|_p = \|w\|_p = 1$ and $\langle j_p(u), w\rangle = 0$. Regard ℓ_p^3 as $\ell_p^2 \oplus_p \mathbb{R}$, with coordinates (ξ, y), where $\xi = (x, z)$. Define

$$T_0(\xi, y) = (yu, -\langle j_p(u), \xi\rangle).$$

Since $\|j_p(u)\|_q = 1$,

$$\|T_0(\xi, y)\|_p^p = |y|^p + |\langle j_p(u), \xi\rangle|^p \le |y|^p + \|\xi\|_p^p.$$

Equality occurs, for example, when $\xi = 0$, so $\|T_0\|_p = 1$.

For any real matrix T, smoothness of the ℓ_p norm gives

$$v_p(T) = \sup_{v \neq 0} \frac{|\langle j_p(v), Tv\rangle|}{\|v\|_p^p}. \tag{935.G1}$$

Set $g_p(s,t) = j_p(s)t - j_p(t)s$. The scalar inequality defining M_p implies

$$|g_p(s,t)| \le M_p(|s|^p + |t|^p) \qquad (s, t \in \mathbb{R}).$$

The numerical form of T_0 is

$$F_0(x, z, y) = g_p(x, cy) + g_p(z, cy).$$

Because $2c^p = 1$, summing the preceding scalar inequalities yields $|F_0(x,z,y)| \le M_p(|x|^p + |z|^p + |y|^p)$. Equality is obtained by choosing both scalar pairs to attain the same signed maximum. Consequently

$$v_p(T_0) = M_p. \tag{935.G2}$$

We next describe all maximizing vectors and their local behavior. A maximizing vector has $y \neq 0$: otherwise $F_0 = 0$ while $M_p > 0$. After a common nonzero real rescaling, use the chart $cy = 1$ and set

$$S = |x|^p + |z|^p + 2.$$

In this chart the normalized numerical form is

$$\phi_0(x,z) = \frac{(1+|x|^p)h_p(x) + (1+|z|^p)h_p(z)}{S}. \tag{935.G3}$$

Every weight is positive. Thus $\phi_0 = M_p$ exactly when

$$x, z \in \{t_p^{-1}, -t_p\},$$

and $\phi_0 = -M_p$ exactly when

$$x, z \in \{t_p, -t_p^{-1}\}.$$

These choices form a finite collection of points. Lemma 1.2 also proves that each maximum of $|\phi_0|$ is nondegenerate. For example, at a positive maximum,

$$M_p - \phi_0(x,z) = \frac{(1+|x|^p)(M_p - h_p(x)) + (1+|z|^p)(M_p - h_p(z))}{S}.$$

Each one-variable deficit has a zero of order exactly two, with positive second derivative. The Hessian of the displayed deficit at a zero is therefore diagonal and positive definite. Negative maxima are handled by replacing h_p with $-h_p$.

Let $t = t_p$ and define

$$\kappa = \frac{1-t^{2p}}{t^2 - t^{2p-2}}, \qquad R = \begin{pmatrix} 1 & \kappa \\ -\kappa & -1 \end{pmatrix}.$$

The denominator is positive, and

$$\kappa - 1 = \frac{(1-t^2)(1+t^{2p-2})}{t^2 - t^{2p-2}} > 0. \tag{935.G4}$$

For real ε set

$$T_\varepsilon(\xi, y) = \big(yu + \varepsilon R\xi, -\langle j_p(u), \xi\rangle\big). \tag{935.G5}$$

Its numerical form is $F_0 + \varepsilon G$, where

$$G(x,z) = |x|^p - |z|^p + \kappa(j_p(x)z - j_p(z)x). \tag{935.G6}$$

This perturbation vanishes at every maximizing vector of T_0. Indeed, their leaf-coordinate ratios are $x/z \in \{1, -t^2, -t^{-2}\}$. The first ratio gives zero directly. For the second ratio,

$$G(-t^2, 1) = t^{2p} - 1 + \kappa(t^2 - t^{2p-2}) = 0.$$

The third follows from antisymmetry under exchanging x, z and homogeneity.

It follows that there are finite constants $C \geq 0$ and $\varepsilon_0 > 0$ (depending on p) such that

$$v_p(T_\varepsilon) \leq M_p + C\varepsilon^2 \qquad (|\varepsilon| \leq \varepsilon_0). \tag{935.G7}$$

Here are the compactness details. On the unit sphere, the maximizing set of $|F_0|$ is finite. In a chart about each such point, nondegeneracy gives $|\phi_0| \leq M_p - c_0\|h\|_2^2$ for some $c_0 > 0$. The normalized perturbation is smooth in that chart, because all coordinates of the maximizer are nonzero, and vanishes at its center. It is therefore bounded in absolute value by $L_0\|h\|_2$. Completing the square gives

$$|\phi_0 + \varepsilon G/S| \leq M_p - c_0\|h\|_2^2 + |\varepsilon| L_0\|h\|_2 \leq M_p + \frac{L_0^2}{4c_0}\varepsilon^2.$$

Outside the finitely many neighborhoods, compactness gives a fixed strict gap below M_p, while the normalized perturbation is bounded. For sufficiently small $|\varepsilon|$ this region contributes at most M_p. Taking the largest of the finitely many constants proves (935.G7).

The norm has a larger leading-order increase. Since $Rw = (1-\kappa)u$ and $\langle j_p(u), w\rangle = 0$,

$$T_\varepsilon(sw, y) = \big((y - (\kappa-1)\varepsilon s)u, 0\big).$$

The input norm is $(|s|^p + |y|^p)^{1/p}$. Taking the norm of this scalar functional on ℓ_p^2 gives

$$\|T_\varepsilon\|_p \geq \big(1 + (\kappa-1)^q|\varepsilon|^q\big)^{1/q}. \tag{935.G8}$$

As $1 < q < 2$, equations (935.G7) and (935.G8) imply

$$\begin{aligned}\frac{v_p(T_\varepsilon)}{\|T_\varepsilon\|_p} &\leq \frac{M_p + C\varepsilon^2}{\big(1 + (\kappa-1)^q|\varepsilon|^q\big)^{1/q}} \\ &= M_p - \frac{M_p}{q}(\kappa-1)^q|\varepsilon|^q + o(|\varepsilon|^q).\end{aligned}$$

The leading correction is strictly negative. Hence every sufficiently small nonzero ε gives $v_p(T_\varepsilon)/\|T_\varepsilon\|_p < M_p$, proving $n(\ell_p^3(\mathbb{R})) < M_p$. □

Dual exponents and the Lebesgue space. For any operator T on a finite-dimensional real normed space, $\|T^*\| = \|T\|$ and $v(T^*) = v(T)$. Indeed, the correspondence between norming pairs (x, x^*) and (x^*, Jx) preserves the scalar $x^*(Tx)$, and finite-dimensional reflexivity makes the correspondence exhaustive. Thus the transpose of a counterexample on ℓ_p^3 gives one on ℓ_q^3, where $q = p/(p-1)$. The substitution $s = t^{p-1}$ in the maximum shows $M_q = M_p$. This proves the strict inequality for $1 < p < 2$ as well.

Finally, for any real 3×3 matrix T, its pointwise action on $L_p([0,1]; \ell_p^3)$ has the same operator norm and numerical radius as T. The upper bounds follow by integrating their pointwise defining inequalities, and constant vector-valued functions give the reverse inequalities. This vector-valued space is isometric to scalar real $L_p([0,1])$ by sending the three coordinates to three equal subintervals and multiplying values by $3^{1/p}$. Consequently $n(L_p([0,1];\mathbb{R})) \leq n(\ell_p^3(\mathbb{R}))$, and the proof of Theorem 1.1 is complete.

1.2 A short explicit certificate at $p = 4$

The perturbation family also gives an explicit algebraic example with a short direct proof, independent of the compactness argument. Put

$$c = 2^{-1/4}, \qquad \varepsilon = 2^{-12}, \qquad T = \begin{pmatrix} \varepsilon & 4\varepsilon & c \\ -4\varepsilon & -\varepsilon & c \\ -c^3 & -c^3 & 0 \end{pmatrix} \quad \text{on real } \ell_4^3, \tag{935.C1}$$

with coordinates ordered as (x, z, y). We prove

$$\frac{v_4(T)}{\|T\|_4} < M_4 = \frac{1}{2\sqrt{2}}.$$

Write $b = cy$, $S = x^4 + z^4 + y^4 = x^4 + z^4 + 2b^4$, and

$$F_0 = b(x^3 + z^3) - b^3(x + z), \qquad G = (x^2 - z^2)(x^2 + z^2 + 4xz).$$

The numerical form of T is $F_0 + \varepsilon G$. Define

$$Q_x = x^2 - \sqrt{2}\,bx - b^2, \qquad Q_z = z^2 - \sqrt{2}\,bz - b^2.$$

Two direct polynomial identities are

$$M_4 S - F_0 = M_4(Q_x^2 + Q_z^2), \tag{935.C2}$$

$$G = (x+z)\big[(x+3z+\sqrt{2}b)Q_x - (3x+z+\sqrt{2}b)Q_z\big]. \tag{935.C3}$$

On $S = 1$, the Euclidean norm of the coefficient vector multiplying (Q_x, Q_z) in (935.C3) is at most

$$|x+z|(4\|(x,z)\|_2 + 2|b|) \le 20^{3/4}. \tag{935.C4}$$

For completeness, let $r = (x^4+z^4)^{1/4}$. Then $|x+z| \le 2^{3/4}$, $\|(x,z)\|_2 \le 2^{1/4}r$, and $2|b| = 2^{3/4}|y|$, while $r^4 + |y|^4 = 1$. Hölder's inequality gives

$$4\|(x,z)\|_2 + 2|b| \le \big((4\,2^{1/4})^{4/3} + (2^{3/4})^{4/3}\big)^{3/4} = (8+2)^{3/4}.$$

This proves (935.C4).

Set $r_Q = (Q_x^2 + Q_z^2)^{1/2}$. Equations (935.C2)–(935.C4) yield

$$\begin{aligned} F_0 + \varepsilon G &\le M_4 - M_4 r_Q^2 + |\varepsilon| 20^{3/4} r_Q \\ &\le M_4 + \frac{20^{3/2}}{4M_4}\varepsilon^2 = M_4 + 20\sqrt{10}\,\varepsilon^2. \end{aligned}$$

Replacing b by $-b$ and ε by $-\varepsilon$ gives the same argument for $-F_0 - \varepsilon G$. Hence

$$v_4(T) \le M_4 + 20\sqrt{10}\,\varepsilon^2. \tag{935.C5}$$

For the norm bound, put $u = (c, c)$ and $w = (c, -c)$. On the isometric copy of ℓ_4^2 consisting of vectors (sw, y),

$$T(sw, y) = \big((y - 3\varepsilon s)u, 0\big).$$

The norm of this scalar functional is therefore a lower bound for the full operator norm:

$$\|T\|_4 \ge \big(1 + 3^{4/3}|\varepsilon|^{4/3}\big)^{3/4}. \tag{935.C6}$$

Let $a = 3^{4/3}2^{-16} \in (0,1)$. The elementary inequality $(1+a)^{3/4} \ge 1 + 3a/8$ on $[0,1]$ follows, for example, by integrating $(3/4)(1+a)^{-1/4} \ge 3/8$. Thus the upper bound in (935.C5) is strictly less than M_4 times the lower bound in (935.C6): after division by M_4 and multiplication by 2^{24}, the required comparison is exactly

$$80\sqrt{5} < 96\,3^{4/3}.$$

It follows already from $80\sqrt{5} < 240 < 288 < 96\,3^{4/3}$. This proves the claimed strict ratio. Pointwise amplification transfers this explicit example to real $L_4([0,1])$.

1.3 An explicit rational certificate at $p = 4$

Theorem 1.3. *For $p \in \{4, 4/3\}$,*

$$n(L_p([0,1];\mathbb{R})) < \frac{1667}{4716} < \frac{1}{2\sqrt{2}} = M_p.$$

The rational upper bound is approximately 0.35347752*, whereas* $1/(2\sqrt{2})$ *is approximately* 0.35355339*.*

1.4 The matrix and the numerical-radius formula

On $\ell_4^3(\mathbb{R})$ define

$$T = \frac{1}{1000}\begin{pmatrix} -10 & 2255 & -40 \\ -1437 & 0 & -1914 \\ 34 & 2480 & 6 \end{pmatrix} = \frac{A}{1000}. \tag{935.1}$$

We will certify

$$v(T) \le \frac{5001}{5000}, \qquad \|T\|_{4\to 4} > \frac{3537}{1250}. \tag{935.2}$$

For $x = (x_1, x_2, x_3) \in \mathbb{R}^3$, set

$$s(x) = x_1^4 + x_2^4 + x_3^4, \qquad F(x) = \sum_{i=1}^{3} x_i^3 (Tx)_i.$$

The unique norming functional of a unit vector of ℓ_4^3 is (x_1^3, x_2^3, x_3^3), so

$$v(T) = \sup_{x \neq 0} \frac{|F(x)|}{s(x)}. \tag{935.3}$$

1.5 Two exact positive Gram matrices

Let

$$z(x) = (x_1^2, x_2^2, x_3^2, x_1x_2, x_1x_3, x_2x_3)^{\mathsf{T}}$$

and define the following symmetric integer matrices:

$$M_+ = \begin{pmatrix} 10102 & -6298 & -14 & -11275 & 200 & 214 \\ -6298 & 10002 & -7757 & 7185 & -4 & 9570 \\ -14 & -7757 & 9942 & -188 & -170 & -12400 \\ -11275 & 7185 & -188 & 12596 & -214 & 4 \\ 200 & -4 & -170 & -214 & 28 & 188 \\ 214 & 9570 & -12400 & 4 & 188 & 15514 \end{pmatrix}, \tag{935.4}$$

$$M_- = \begin{pmatrix} 9902 & -6433 & -8 & 11275 & -200 & 218 \\ -6433 & 10002 & -7646 & -7185 & -4 & -9570 \\ -8 & -7646 & 10062 & -204 & 170 & 12400 \\ 11275 & -7185 & -204 & 12866 & -218 & 4 \\ -200 & -4 & 170 & -218 & 16 & 204 \\ 218 & -9570 & 12400 & 4 & 204 & 15292 \end{pmatrix}. \tag{935.5}$$

Direct expansion gives the exact polynomial identities

$$\frac{1}{10000} z(x)^{\mathsf{T}} M_+ z(x) = \frac{5001}{5000} s(x) - F(x), \tag{935.6}$$

$$\frac{1}{10000} z(x)^{\mathsf{T}} M_- z(x) = \frac{5001}{5000} s(x) + F(x). \tag{935.7}$$

For example, the coefficient of $x_1^3x_2$ in the numerator on the left of (935.6) is $2(-11275) = -22550 = -10A_{12}$. All coefficients of absent monomials cancel.

Both matrices are positive definite. Their leading principal minors, in orders one through six, are the following positive integers:

$$\begin{aligned} M_+ : 10102, \quad & 61375400, \quad 976429002, \\ & 6319567977, \quad 39156751988, \quad 156153016840; \end{aligned} \tag{935.8}$$

$$M_- : 9902, \quad 57656315, \quad 466262882, \\ 4413501692, \quad 8528707160, \quad 32937577888. \tag{935.9}$$

Sylvester's criterion and (935.6)–(935.7) therefore imply $|F(x)| \le (5001/5000)s(x)$ for every real x. Equation (935.3) proves the numerical-radius bound in (935.2).

1.6 An exact norm witness and comparison with M_4

For $w = (185, 1000, -179)^{\mathsf{T}}$ one has

$$Aw = (2260310, 76761, 2485216)^{\mathsf{T}},$$

$$\sum_i w_i^4 = 1002197976306, \qquad \sum_i (Aw)_i^4 = 64248592946651712875988977.$$

The integer comparison

$$625\sum_i (Aw)_i^4 - 14148^4 \sum_i w_i^4 = 9164935582748540323065 29 \\ > 0 \tag{935.10}$$

is exactly $\|Tw\|_4^4 > (3537/1250)^4 \|w\|_4^4$, because $14148/5 = 1000(3537/1250)$. This proves the second bound in (935.2) and hence

$$\frac{v(T)}{\|T\|_{4\to 4}} < \frac{5001/5000}{3537/1250} = \frac{1667}{4716}. \tag{935.11}$$

For $0 \le t \le 1$ the identity

$$1 + t^4 - 2\sqrt{2}(t - t^3) = (t^2 + \sqrt{2}t - 1)^2 \ge 0$$

has equality at $t = (\sqrt{6} - \sqrt{2})/2$. Therefore

$$M_4 = \max_{0\le t\le 1} \frac{t - t^3}{1 + t^4} = \frac{1}{2\sqrt{2}}.$$

Finally, $4716^2 - 8 \cdot 1667^2 = 9544 > 0$, so $1667/4716 < 1/(2\sqrt{2})$. Thus (935.11) disproves the corresponding three-dimensional formula.

1.7 Pointwise amplification to the exact Lebesgue space

Set $\mathcal{X} = L_4([0,1]; \ell_4^3(\mathbb{R}))$ and define $(\mathcal{T}f)(t) = T(f(t))$. Pointwise integration gives $\|\mathcal{T}\| \le \|T\|_{4\to 4}$, while constant vector-valued functions give the reverse inequality. Thus the norms are equal. For a unit vector $f \in \mathcal{X}$, its norming functional is pointwise (f_1^3, f_2^3, f_3^3). Integrating (935.6)–(935.7) gives

$$\left| \int_0^1 \sum_i f_i(t)^3 (\mathcal{T}f)_i(t)\, dt \right| \le \frac{5001}{5000} \int_0^1 \sum_i |f_i(t)|^4\, dt.$$

Consequently $v(\mathcal{T}) \le 5001/5000$; indeed constant functions show $v(\mathcal{T}) = v(T)$.

The explicit surjective isometry from $\mathcal{X}$ onto the scalar real Lebesgue space is

$$(Uf)(s) = 3^{1/4} f_i(3s - i + 1), \qquad \frac{i-1}{3} \le s < \frac{i}{3}, \quad i = 1, 2, 3.$$

Transport $\mathcal{T}$ through U and normalize by its operator norm. Equation (935.11) now proves the assertion of Theorem 1.3 for the precise space in the catalogue. This is a pointwise amplification, not a conditional-expectation compression; the latter would not automatically preserve the numerical radius.

1.8 The dual exponent $4/3$

For an operator R on a reflexive real Banach space X, $\|R^*\| = \|R\|$ and $v(R^*) = v(R)$. To verify the latter identity, every norming pair (x, x^*) for X gives the norming pair $(x^*, J_X x)$ for X^*, with

$$(J_X x)(R^* x^*) = x^*(Rx).$$

Reflexivity supplies the converse correspondence for every norming pair in $X^* \times X^{**}$. Apply this to $X = \ell_4^3$. The matrix T^{T} on $\ell_{4/3}^3$ has the same norm and numerical radius as T. Pointwise amplification works for every $1 < p < \infty$: the norming functional is $\operatorname{sign}(f_i)|f_i|^{p-1}$, and the finite-dimensional numerical-radius inequality integrates with degree p. Hence (935.11) also applies on real $L_{4/3}([0,1])$. The substitution $s = t^3$ in the defining maximum gives $M_{4/3} = M_4$.

Verification and scope. The displayed integers form a complete certificate. The supplied `verify935_exact.py` checks both polynomial identities, all twelve principal minors, (935.10), and the strict constant comparison using only exact integer and rational arithmetic. An independent verifier uses separate formulas for the fifteen quartic coefficients and rational LDL^{T} decomposition. Floating-point searches were used only to discover the certificate, and are not premises of the proof. The two-dimensional theorem of [1] is not contradicted: this matrix has dimension three. The counterexample refutes the real-case conjecture in [2, Problem 5.1] and resolves the exact catalogue identity question [3]; it does not determine the numerical index itself.

1.9 A certified interval of exponents

The same matrix yields a quantitative neighborhood of counterexamples.

Proposition 1.4. *The identity in Problem 935 fails for every exponent in either interval*

$$\left[\frac{39999}{10000}, \frac{40001}{10000}\right], \qquad \left[\frac{40001}{30001}, \frac{39999}{29999}\right].$$

Proof. Write v_p and $\|\cdot\|_p$ for the numerical radius and operator norm on real ℓ_p^3, and keep the matrix T from (935.1). Set

$$\varepsilon = \frac{1}{10000}, \quad \gamma = \frac{5001}{5000}, \quad L = \frac{3537}{1250}, \quad R = \|T\|_{\infty\to\infty} = \frac{3351}{1000}.$$

We first treat $|p-4| \le \varepsilon$. For a nonzero real vector x, use homogeneity to arrange $\max_i |x_i| = 1$, and put

$$S_p(x) = \sum_i |x_i|^p, \qquad F_p(x) = \sum_i \operatorname{sign}(x_i)|x_i|^{p-1}(Tx)_i.$$

Then $S_p(x) \ge 1$, and at least one coordinate has absolute value one. For $a > 0$,

$$\max_{0\le t\le 1} t^a |\log t| = \frac{1}{ea},$$

with the value at zero understood by continuity. The mean value theorem, $p \ge 39/10$, and $|(Tx)_i| \le R$ therefore imply

$$|F_p(x) - F_4(x)| \le \frac{2R}{e(29/10)}\varepsilon, \qquad |S_p(x) - S_4(x)| \le \frac{2}{e(39/10)}\varepsilon. \tag{935.12}$$

The factor two is valid because the summand corresponding to a coordinate of absolute value one vanishes in each difference. The certificate already proved $|F_4(x)| \le \gamma S_4(x)$. Since $e > 8/3$,

it follows that

$$|F_p(x)| \le \gamma S_p(x) + C\varepsilon,$$
$$C = \frac{15}{2}\left(\frac{R}{29} + \frac{\gamma}{39}\right) = \frac{399237}{377000}.$$

After division by $S_p(x) \ge 1$ and taking the supremum,

$$v_p(T) \le \gamma + C\varepsilon. \tag{935.13}$$

Standard comparison of the three-dimensional ℓ_p norms gives

$$\|T\|_p \ge 3^{-d}\|T\|_4, \qquad d = \left|\frac{1}{p} - \frac{1}{4}\right| \le \frac{\varepsilon}{4p}.$$

For $0 \le d \le 1$, convexity gives $3^d \le 1 + 2d$. As $p \ge 39/10 > 7/2$,

$$3^d \le 1 + \frac{\varepsilon}{2p} \le 1 + \frac{\varepsilon}{7}.$$

The strict bound $\|T\|_4 > L$ hence yields

$$\frac{v_p(T)}{\|T\|_p} < \frac{(\gamma + C\varepsilon)(1 + \varepsilon/7)}{L} = \frac{87994832581079}{248910480000000}. \tag{935.14}$$

For $p > 2$, put $f_p(t) = (t - t^{p-1})/(1 + t^p)$ on $[0, 1]$. Direct differentiation gives

$$\frac{\partial f_p(t)}{\partial p} = \frac{(-\log t)t^{p-1}(1 + t^2)}{(1 + t^p)^2}, \qquad \left|\frac{\partial f_p(t)}{\partial p}\right| \le \frac{2}{e(p-1)} \le \frac{15}{58}$$

throughout the interval under consideration. Evaluate f_p at the maximizing point $t_0 = (\sqrt{6} - \sqrt{2})/2$ of f_4. Since $8 \cdot 7071^2 = 399992328 < 400000000 = 20000^2$,

$$M_p \ge f_p(t_0) \ge M_4 - \frac{15}{58}\varepsilon > \frac{7071}{20000} - \frac{15}{58}\varepsilon = \frac{51261}{145000}. \tag{935.15}$$

The two rational bounds are strictly separated:

$$\frac{51261}{145000} - \frac{87994832581079}{248910480000000} = \frac{1030282921}{248910480000000} > 0.$$

Thus $v_p(T)/\|T\|_p < M_p$ throughout the first interval. Pointwise amplification gives the same strict inequality for the numerical index of real $L_p([0, 1])$.

For the conjugate exponent $q = p/(p-1)$, reflexive duality gives $v_q(T^{\mathsf{T}}) = v_p(T)$ and $\|T^{\mathsf{T}}\|_q = \|T\|_p$. The substitution $s = t^{p-1}$ in the maximum defining M_q gives $M_q = M_p$. The decreasing map $p \mapsto p/(p-1)$ carries the first interval onto the second, proving that case as well. □

References

[1] R. Chiclana, *The Numerical Index of Two-Dimensional Real ℓ_p Spaces*, preprint, June 2026. arXiv:2606.01675.

[2] M. Martín, J. Merí, and M. Popov, *On the numerical radius of operators in Lebesgue spaces*, J. Funct. Anal. **261** (2011), 149–168. doi:10.1016/j.jfa.2011.03.007. Problem 5.1.

[3] MColbrook/AIM, *AIM 935: The numerical index of real Lebesgue spaces*, repository statement at revision `5d7e26e8444e`. https://github.com/MColbrook/AIM/blob/5d7e26e8444eb98de91988454564e45e011107c4/problems/935-the-numerical-index-of-real-lebesgue-spaces.md.